

Integration of Sysmon with Wazuh to Detect Fileless Malware

1- Definition

By integrating Sysmon (System Monitor) with Wazuh, we can detect fileless malware, like malicious PowerShell scripts or memory-based attacks.

PowerShell scripts are text files containing a series of commands built into Windows. While IT administrators use them to automate routine tasks, attackers use them because they are already trusted tools on the system and can run commands without needing to install extra software.

Memory-based attacks are cyberattacks that run entirely inside your computer's temporary memory (RAM) instead of saving files to the hard drive. Because no files are ever written to the disk, traditional antivirus programs often fail to notice them.

2- Centralized Group Configuration

We are using centralized group configuration, and for this case we will use windows group, So when we push changes from our Wazuh manager, it will apply those changes to every agent.

3- Downloading Sysmon and Configuration Files

First: we need to download Sysmon on <https://learn.microsoft.com/en-us/sysinternals/downloads/sysmon>

After that, we need to download the Olaf Hems XML configuration file on <https://github.com/tim87tchoua/sysmon-modular/blob/master/sysmonconfig.xml> So, click on download button

4- Preparing the Installation Files

Now we have downloaded both files: **Sysmon and sysmonconfig.xml**

extract Sysmon, And paste the XML file into the Sysmon folder.

After that, move the Sysmon folder to the C drive.

Go into the folder, copy the path (C:\Sysmon)

5- Installing Sysmon

Open Windows PowerShell, and type `cd C:\Sysmon` then hit enter.

After that, Type `.\Sysmon64.exe -accepteula -l sysmonconfig.xml` and hit enter.

Our Sysmon will be installed on the endpoint.

6- Configuring Wazuh Agent Log Collection

Next, navigate to the Wazuh dashboard (menu → Agent Management → group → Windows → agent.conf to past this configuration block:

```
<!-- Sysmon Log Collection -->
```

```
<localfile>
```

```
<location>Event Channel Name Here</location> (Microsoft-Windows-Sysmon/Operational)
```

```
<log_format>eventchannel</log_format>
```

```
</localfile>
```

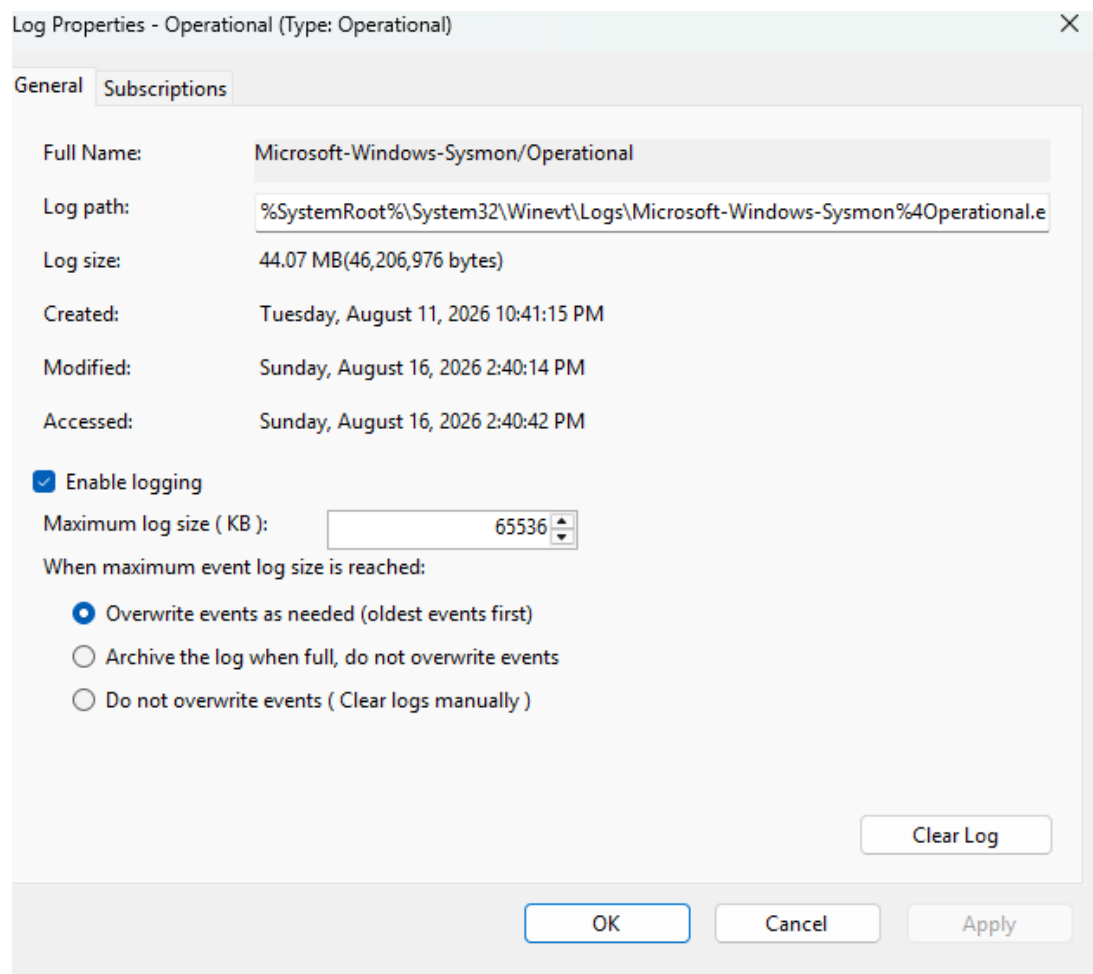
Here we will define our Sysmon channel name.

Save the configuration and restart the Wazuh manager.

7- Locating the Event Channel Name

Question: Where can we find that Event Channel Name?

In the Windows search bar, type Event Viewer, navigate to Application and Services logs → Microsoft → Windows → Sysmon → Operational, right click → properties to see this:



8- Testing and Verification

Go back to our windows agent, run PowerShell as administrator and execute this powershell command:

```
PS C:\Windows\system32> powershell.exe -nop -w hidden -c "Write-Host 'Testing Fileless Detection'"
```

Return to the Wazuh dashboard to confirm that alerts are successfully triggering and capturing the activity.



WAZUH
Open Source Security
Monitoring Platform

Integration of Sysmon with Wazuh to Detect Fileless Malware

Detect Threats Before They Leave No File Behind

Collect Analyze Detect Respond Stay Secure



**Sysinternals
Sysmon**

Microsoft

IT Runs. We Monitor.
You Stay Safer.

1 Definition & Overview

By integrating Sysmon (System Monitor) with Wazuh, we can detect fileless malware, like malicious PowerShell scripts or memory-based attacks.



What is PowerShell?

PowerShell scripts are text files containing a series of commands built into Windows. While IT administrators use them to automate routine tasks, attackers use them because they are already trusted tools on the system and can run commands without needing to install extra software.



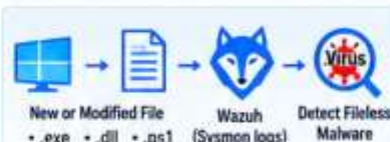
What are Memory-based attacks?

Memory-based attacks run entirely in your computer's temporary memory (RAM), instead of saving files to the hard drive. Because no files are written to disk, traditional antivirus programs often fail to notice them.

"Detect what others miss!"

2 Why This Integration is Important

- Automatic threat hunting (no manual checking).
- Real-time alerts if a malicious file is detected.
- Identify the user and process behind suspicious activity.
- Helps with compliance requirements (PCI DSS, GDPR).
- Improves overall endpoint security.

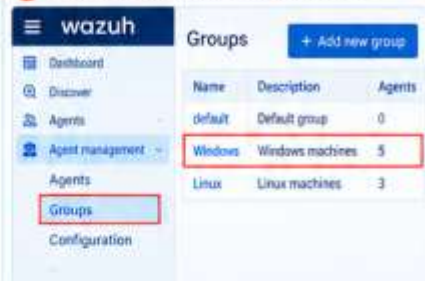


Wazuh + Sysmon = Deep Visibility
Even When There is No File on Disk

3 Centralized Group Configuration

We use centralized group configuration (Windows group). When we push changes from the Wazuh manager, it will apply those changes to every Windows agent.

- Log in to Wazuh dashboard.
- Go to **Agent Management** → **Groups**.
- Select your Windows group (or create one).
- Edit the agent configuration file (agent.conf).



4 Download Sysmon and Configuration Files

- Download Sysmon from Microsoft Sysinternals:
<https://www.microsoft.com/en-us/sysinternals/downloads/sysmon>

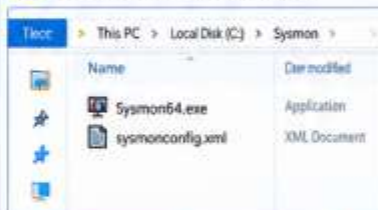


- Download the Olaf Hems XML configuration file:
<https://github.com/tim87tchoua/sysmon-modular/blob/master/sysmonconfig.xml>



5 Prepare the Installation Files

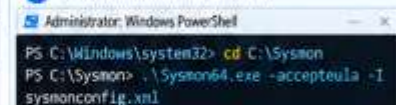
- Extract the Sysmon zip file.
- Copy the downloaded sysmonconfig.xml into the Sysmon folder.
- Move the Sysmon folder to the C: drive (C:\Sysmon).
- Open the folder and copy the full path.



Folder path: C:\Sysmon

6 Install Sysmon on Windows Agent

- Open Windows PowerShell as Administrator.
- Navigate to the Sysmon folder.
- Run the installation command.



System Monitor v15.21 - System activity monitor
Copyright (C) 2014-2024 Microsoft Corporation
Sysmon64 installed.
SysmonDrv installed.
Starting SysmonDrv.
SysmonDrv started.
Starting Sysmon64.
Sysmon64 started.

✓ Sysmon successfully installed on the endpoint!

7 Configure Wazuh Agent Log Collection

- In the Wazuh dashboard, go to **Agent Management** → **Groups** → **Windows** → open agent.conf.
- Add the following configuration block:

```

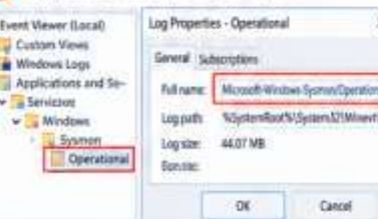
<!-- Sysmon Log Collection -->
<localfile>
  <location>Microsoft-Windows-Sysmon/Operational</location>
  <log_format>eventchannel</log_format>
</localfile>
  
```

This tells Wazuh to collect Sysmon logs from the Microsoft-Windows-Sysmon/Operational channel.

- Save the configuration and restart the Wazuh manager.

8 Find the Event Channel Name

- On the Windows agent, open Event Viewer.
- Go to **Application and Services Logs** → **Microsoft** → **Windows** → **Sysmon** → **Operational**.
- Right-click on **Operational** → **Properties**.
- Copy the Full Name (channel name).



9 Create a Local Rule (local_rules.xml)

- On the Wazuh manager, open the rules file:
`sudo nano /var/ossec/etc/rules/local_rules.xml`
- Add this rule at the end:

```

<group name="sysmon,virustotal">
  <!-- This rule will fire when a new .exe or .dll or .ps1 file -->
  <rule id="100108" level="7">
    <if_sid>10005</if_sid>
    <field name="win.eventdata.targetfilename">
      \\.\\.*\\.dll|\\.ps1$
    </field>
    <description>File - New executable or script created -> VirusTotal Scan Triggered</description>
  </rule>
</group>
  
```

✓ Save the file.

10 Restart Wazuh Manager

- On the Wazuh server, run the command:
`sudo systemctl restart wazuh-manager.service`
- This applies the new configuration to all agents.

11 Test Fileless Malware Detection

- On the Windows agent, open PowerShell as Administrator and run a test command:
`PS C:\Windows\system32> powershell.exe -nop -w hidden -c "Write-Host 'Testing Fileless Detection'"`
- Go to the Wazuh dashboard (**Security events**) to confirm that alerts are triggered.

12 Simulate an Attack with EICAR

- Download the EICAR test file: <https://www.eicar.org/download-anti-malware-testfile/>
 - Save the test file in a folder you are monitoring with FIM.
 - Check Wazuh SIEM → **Security events**.
 - You should see an alert, and VirusTotal will detect the malware.
- ✓ Malware detected! VirusTotal integration is working.



WAZUH
Open Source Security
Monitoring Platform



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"Knowledge
Builds a Safer
Tomorrow"